Construction Chemicals

Product Datasheet
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Protective Coating

O-Prime ST

Surface-Tolerant Epoxy Primer
for Steel, Light Metals & Concrete



Product Description

O-Prime ST is a two-component, solvent-based, surface-tolerant epoxy primer. It gives reliable adhesion on steel prepared by hand or power tools (St 2) as well as blast-cleaned steel (Sa 2½), and is suitable for concrete, light metals and their alloys. It is the base primer of the O MIX protective coating systems.

Uses

- Primer for structural steel, tanks, pipes and steel structures.
- Maintenance painting where only hand or power-tool cleaning is possible (St 2).
- Primer for light metals and their alloys.
- Primer on concrete before epoxy and polyurethane protective systems.
- Base coat in O MIX Acid Lock, Potable Water Tank and PU Protective systems.

Advantages

- Surface tolerant: St 2 or Sa 2½.
- High solids 85 %, high build.
- Excellent adhesion to steel and concrete.
- Wide DFT range 100 – 1000 µm.
- Compatible with epoxy and PU topcoats.

Application

1. **Surface Preparation**
   Remove rust, mill scale, oil and loose paint to St 2 minimum (Sa 2½ preferred).
2. **Mixing**
   Add Part B to Part A (4 : 1 by weight) and mix 2 – 3 min with a slow-speed mixer.
3. **Application**
   Apply by brush, roller or airless spray at 0.18 – 0.20 kg/m² (100 µm DFT).

Consumption

0.18 – 0.20 kg/m²
at 100 µm DFT — theoretical 8.5 m²/L
Practical consumption depends on surface profile, method and losses.

Packaging

20 kg sets (Part A 16 kg + Part B 4 kg) and 5 kg sets (Part A 4 kg + Part B 1 kg) in metal pails.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight, heat and ignition sources.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	4 : 1 by weight	—
Finish	Semi-gloss	—
Colours	Grey, red, white	—
Solids content	85 %	—
Recommended DFT	100 – 1000 µm	—
Theoretical coverage	8.5 m²/L at 100 µm DFT	—
Pot life (25 °C)	2 hours	—
VOC	316 g/L	—
Flash point	27 °C	—
Min. substrate temperature	+10 °C	—
Shelf life	12 months	—

Consumption

DFT	Theoretical coverage	Consumption
100 µm	8.5 m²/L	0.18 – 0.20 kg/m²
150 µm	5.7 m²/L	0.27 – 0.30 kg/m²
200 µm	4.25 m²/L	0.36 – 0.40 kg/m²

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Substrate temperature: min. +10 °C.
- Substrate at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Steel: St 2 minimum; Sa 2½ for immersion or severe exposure.
- Concrete: at least 28 days old, sound, dry (< 4 % moisture) and free of laitance.

Application Notes

- Mix complete sets only; partial mixing must be weighed exactly at 4 : 1.
- Apply by brush, roller or airless spray; use brush for stripe coats on edges, welds and bolts.
- Overcoat with the specified O MIX epoxy or PU coat within the recoat window stated for the system.
- Thin only with the recommended epoxy thinner where needed for spray.
- Do not apply on wet surfaces or when condensation is expected.

Health & Safety

- Solvent-based: flammable. Keep away from heat, sparks and open flames; no smoking.
- Wear chemical-resistant gloves, safety goggles and overalls; use a respirator with organic vapour cartridges in enclosed spaces.
- Uncured epoxy may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation during application and curing.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean tools and equipment immediately with epoxy thinner.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.